

Googlantir
A Zero-Shot Time-Series Forecasting Application
Based on Google Research TimesFM Architecture

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Abstract

This report presents Googlantir, a web-based application designed for financial time-series forecasting. The system leverages TimesFM, a foundation model developed by Google Research, to generate probabilistic predictions without requiring domain-specific training or fine-tuning. The application provides an accessible interface for exploring zero-shot forecasting capabilities across multiple temporal granularities, offering configurable context windows and forecast horizons with quantified uncertainty estimates.

1 Introduction

Googlantir is a forecasting tool implemented as a Streamlit-based web application in Python. The system is designed to perform predictive analysis on financial time series data retrieved from public markets via the Yahoo Finance API. The primary distinguishing characteristic of this implementation is its reliance on TimesFM (Time Series Foundation Model), a pre-trained transformer-based architecture capable of zero-shot inference.

The zero-shot capability eliminates the conventional requirement for model retraining or fine-tuning on target-specific datasets. This property enables immediate deployment across heterogeneous time series without incurring the computational overhead or data requirements typically associated with supervised learning approaches. The model operates effectively in a local execution environment, requiring no external API calls for inference once the pre-trained weights are loaded.

2 System Architecture

The application comprises three primary components structured in a modular architecture:

2.1 User Interface Layer

The front-end is implemented using Streamlit, a Python framework for rapid web application development. The interface provides:

- Ticker symbol input with validation
- Three operational modes corresponding to different temporal resolutions (hourly, daily, weekly)
- Interactive parameter controls for context length and forecast horizon
- Real-time visualization of predictions with confidence intervals

The design incorporates educational components explaining the tradeoffs inherent in time-series forecasting, particularly the noise-bias balance in context window selection.

2.2 Data Acquisition Module

The data layer utilizes the `yfinance` library to retrieve historical price data from Yahoo Finance. The module handles:

- Multi-granularity data fetching (1-hour, 1-day, 1-week intervals)
- Timezone normalization for cross-market compatibility
- Differentiation between cryptocurrency (24/7) and equity (trading hours) data streams
- Validation of data sufficiency relative to model requirements

2.3 Forecasting Engine

The core prediction logic is encapsulated in a separate module that interfaces with the TimesFM model. Key operations include:

- Model instantiation with configurable context and horizon parameters
- Input normalization using standardization (zero mean, unit variance)
- Inference execution with quantile regression for uncertainty quantification
- Post-processing to generate point forecasts and confidence bands (10th, 30th, 70th, 90th percentiles)

The model configuration is cached using Streamlit's resource management to avoid redundant recompilation across sequential forecasts.

3 Technical Implementation Details

3.1 TimesFM Model Characteristics

TimesFM is a decoder-only transformer architecture pre-trained on approximately 100 billion real-world and synthetic time-series data points. The model exhibits several properties relevant to this application:

- **Zero-shot generalization**: The model performs inference on unseen time series without gradient updates or retraining. This capability stems from its pre-training on a diverse corpus spanning multiple domains, frequencies, and statistical characteristics.
- **Local execution**: The model operates entirely within the local computational environment. Once the 200-million-parameter checkpoint is downloaded, no external network requests are required for inference.
- **Frequency-agnostic processing**: The architecture accepts a frequency token as input, enabling consistent handling of time series with different sampling rates (hourly, daily, weekly) without structural modifications.
- **Probabilistic outputs**: Beyond point predictions, the model generates a continuous quantile function, allowing for uncertainty estimation at arbitrary confidence levels.

3.2 Challenges and Solutions

3.2.1 Temporal Discontinuities in Financial Data

Financial markets exhibit non-uniform time structures due to trading hours and weekends. For hourly and daily resolutions, naive date generation would introduce artificial gaps. The solution implemented involves:

- Conditional logic distinguishing cryptocurrency (continuous) from equity (discrete) markets
- Filtering of non-trading hours (pre-market and after-hours periods)
- Weekend exclusion for equity markets
- Plotly range breaks to prevent visual distortion in charts

3.2.2 Data Availability Constraints

The Yahoo Finance API imposes implicit rate limits and historical data constraints. The application mitigates these through:

- Intelligent period mapping (e.g., "2 weeks" $\rightarrow$ 1-month fetch with post-processing truncation)
- Pre-validation of data sufficiency (minimum 35 observations) before model invocation
- Informative error messages distinguishing ticker invalidity from insufficient historical depth

3.2.3 Model Recompilation Overhead

The TimesFM implementation requires JIT compilation when context or horizon parameters change. To maintain interface responsiveness:

- Model instances are cached using `@st.cache_resource` decorator
- Cache keys incorporate both context length and forecast horizon
- Initial compilation latency is communicated via loading indicators

4 Methodological Considerations

4.1 Context-Horizon Tradeoff

The application enforces no hard constraints on parameter selection but provides guidance through educational components. The fundamental tradeoff is:

- **Short context**: High sensitivity to recent dynamics, increased variance, reduced ability to capture long-term periodicity
- **Long context**: Better seasonality detection, reduced noise sensitivity, potential staleness if underlying regime has shifted

The recommended heuristic of maintaining a 3:1 to 4:1 context-to-horizon ratio emerges from empirical observations in time-series forecasting literature, balancing information content against overfitting risks.

4.2 Uncertainty Quantification

The confidence intervals presented are derived from the model's learned quantile function. The interpretation requires caution:

- Intervals represent in-sample calibration on the pre-training distribution
- Out-of-distribution scenarios (e.g., market crashes) may exhibit poor calibration
- Wider intervals indicate regions where the model extrapolates beyond training support

5 Limitations and Future Work

The current implementation exhibits several constraints:

- **Univariate focus**: The model processes only closing prices, ignoring volume, volatility, and other market signals
- **No exogenous variables**: Macroeconomic indicators, news sentiment, and earnings data are not incorporated
- **Static model**: The frozen weights cannot adapt to regime changes without explicit retraining on updated data
- **Computational requirements**: The 200M-parameter model requires approximately 1GB RAM; larger contexts increase memory footprint

Potential extensions include multi-variate forecasting support, integration of external data sources, and ensemble methods combining TimesFM with domain-specific heuristics.

6 Conclusion

Googlantir demonstrates the practical applicability of foundation models to financial time-series forecasting in a zero-shot regime. The implementation prioritizes accessibility and interpretability, providing both technical users and domain practitioners with tools to explore model behavior across different configurations. The modular architecture facilitates extension and experimentation, while the educational components address common misconceptions regarding forecasting methodology.

The zero-shot capability of TimesFM represents a significant reduction in deployment complexity compared to traditional supervised learning pipelines, eliminating data collection, labeling, and model training phases. This property, combined with local execution, positions the system as a viable tool for rapid prototyping and educational exploration of time-series forecasting principles.

Repository: `https://github.com/npasi/Googlantir`
Technologies: Python, Streamlit, PyTorch, TimesFM, yfinance, Plotly